P-S-560B
March 14, 1975
SUPERSEDING
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April 22, 1964

FEDERAL SPECIFICATION

SOAP, FLOOR, LIQUID AND PASTE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of floor soap.

1.2 Classification. The floor soap covered by this specification shall be of one grade, of the following types as specified (see 6.1):

Type I - Liquid.
Type II - Paste.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-705 - Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-704 - Pails, Metal: Shipping, Steel (1 Through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc, Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society For Testing and Materials (ASTM) Standards:

D 92 - Flash and Fire Points by Cleveland Open Cup.
D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. The soap shall be a uniform liquid made solely from potash and either whole neutral vegetable oils, distilled vegetable-oil fatty acids or distilled vegetable-oil fatty acids and rosin acids. The soap shall conform to the requirements set forth in this section.

3.1.2 Type II. The soap shall be a firm, uniform translucent gel or paste made solely from potash and either whole neutral vegetable oils, distilled vegetable-oil fatty acids or distilled vegetable-oil fatty acids and rosin acids (see 6.4). The soap shall conform to the requirements set forth in this section.

3.2 Composition. The composition of the soaps shall conform to the requirements specified in table I when tested as specified in ASTM D 460.

3.3 Solubility and sudsing. The soaps shall be readily and completely dissolved in distilled water at 15.5° to 20°C. (60° to 68°F.) to give a 0.12 to 0.15 percent solution. This solution shall yield at least 150 ml of suds when tested in accordance with 4.3.5.1.

3.4 Flash point. The soaps shall have flash points above their boiling points when tested in accordance with 4.3.5.2.

3.5 Stability. The soaps shall not become rancid or separate or otherwise deteriorate when tested in accordance with 4.3.5.3.

3.6 Color. Type I soap, as received, and 20 percent water solutions of type II soaps shall be not darker than number 17 of the Gardner Liquid Color Scale when tested in accordance with 4.3.5.4.

3.7 Computation for type II. The percentage of moisture shall be computed and reported by the testing laboratory on the soap as received. The percentage of all other constituents shall be calculated and reported on an assumed moisture content of 50 percent. To make calculations, multiply the percentages of constituents other than

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moisture by a factor (F) derived as follows:

$$F = \frac{50}{100 - \text{percent moisture}}$$

TABLE I. Composition

	Type I		Type II	
	Percent by weight		Percent by weight	
	Minimum	Maximum	Minimum	Maximum
Moisture (toluene distillation method)	-	-	-	55.0
Total matter insoluble in alcohol	-	0.5	-	1.0
Free alkali, calculated as potassium hydroxide (KOH)	-	0.05	-	0.1
Free acid, calculated as oleic acid	-	0.1	-	0.2
Alkaline salts, calculated as potassium carbonate (K_2CO_3)	-	0.1	-	0.2
Matter insoluble in distilled water	-	0.1	-	0.2
Chloride, calculated as potassium chloride (KCl)	-	0.3	-	0.5
Unsaponified and unsaponifiable matter	-	0.8	-	1.8
Anhydrous soap, calculated as potash soap	20.0	-	43.0	-
Total sodium and potassium oxides	-	0.2	-	0.5
Glycerol	-	1.8	-	4.0
Rosin acids, percent	-	1.0	-	2.0
Sugar	-	none	-	none
			Numerical values	
Iodine number (Wijs) of mixed fatty acids derived from the soap	80	150	80	150
Acid number of mixed fatty acids derived from the soap	190	205	190	205

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Inspection shall be in accordance with provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling, and inspection lot for examinations and tests shall consist of soap of a single type and class made with the same components during a normal production run and submitted for inspection and tests at one time.

4.2.1 Inspection of materials and components. The supplier is responsible for insuring that all materials and components used in the end item were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.4. Random samples shall be drawn from each lot of end items offered for inspection. The lot size for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of shipping containers for examination under 4.3.1 and in units of primary containers of the same specified capacity, consisting of soap of the same specified type, for examination in 4.3.2 and 4.3.3.

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4.3.1 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.1.1 Examination of closure, waterproofing and banding of containers. When shipping containers are required to comply with PPP-B-636, or PPP-P-704, the examination for defects in closure, waterproofing and banding shall be in accordance with appendix of that specification.

4.3.2 Examination of the soap product for defects in form and workmanship. The sample unit for this examination shall be the contents of one filled primary container.

Examine	Defect
Form:	
Type I	Not liquid.
Type II	Not paste or firm translucent gel.
Workmanship	Dirt.
	Foreign matter.
	Not homogeneous.

4.3.3 Examination for defects in net contents. The sample unit for this examination shall be one filled primary container. The average net contents per container shall be not less than the specified volume or weight, as applicable.

4.3.4 Inspection levels and acceptable quality levels for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1	S-2	4.0
4.3.2	S-4	2.5
4.3.3	S-2	-

4.3.5 Testing of the end item. The end item shall be tested for the characteristics listed in table III. The inspection level for determining the sample size shall be S-1. The sample unit for testing shall be a composite of samples selected. The quantity of sample drawn from each container shall be one pint for type I and one pound for type II. The composite for testing shall be one pint for type I and one pound for type II. All samples shall be drawn and composited in accordance with ASTM D-460.

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TABLE I.I. Instructions for testing

Characteristic	Specification reference: Requirement	Specification reference: Test method (paragraphs)	Requirements applicable to Composite unit	Number determinations per composite sample	Results reported as: Pass or Fail 2/	Results reported as: Numerically to nearest 3/
Moisture (type II only)	Table I	12-15 (Method B)		Average of 2		0.5 percent
Total matter insoluble in alcohol	Table I	16-17		Average of 2		0.1 percent
Free alkali, calculated as potassium hydroxide (KOH)	Table I	18		Average of 2		0.01 percent
Free acid, calculated as oleic acid	Table I	18		Average of 2		0.1 percent
Alkaline salts, calculated as potassium carbonate (K_2CO_3)	Table I	20		Average of 2		0.1 percent
Matter insoluble in distilled water	Table I	19		Average of 2		0.1 percent
Chloride, calculated as potassium chloride (KCl)	Table I	50-52		Average of 2		0.1 percent
Unsaponified plus unsaponifiable matter	Table I	30-32		Average of 2		0.1 percent
Anhydrous soap, calculated as potash soap	Table I	21-22		Average of 2		0.1 percent
Total sodium and potassium oxides	Table I	23-26		Average of 2		0.1 percent
Glycerol	Table I	79-83		Average of 2		0.1 percent
Iodine number	Table I	47-49		Average of 2		Number
Rosin acids	Table I	37-40	X	1 and Average of 2	X	0.1 percent Number
Sugar	Table I	86-87	X	1	X	
Acid number	Table I	45-46		Average of 2		Number
Solubility and sudsing	3.3	4.3.5.1	X	1	X	
Flash point	3.4	4.3.5.2	X	1	X	
Stability	3.5	4.3.5.3	X	1	X	
Color	3.6	4.3.5.4	X	1	X	

1/ All tests except solubility and sudsing, flash point, stability, and color are performed in accordance with ASTM D 460.
2/ If failure is indicated, report description of any failures.
3/ Test reports shall include all values on which results are based.

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4.3.5.1 Solubility and sudsing. Dissolve 4.00 g of type I or 2.00 g of type II in 500 ml of distilled water at 18° ± 2°C (64 ± 3.8°F). Place 50 ml of the test solution in a 250 ml glass stoppered graduated cylinder and shake by inverting the cylinder 30 times in the space of 60 seconds. Put the cylinder down, remove the stopper and read the total volume of foam after 5 seconds.

4.3.5.2 Flash point. The flash point of soap samples shall be determined in accordance with ASTM D 92.

4.3.5.3 Stability. Place 100 g of soap sample in a clean screw cap bottle, and place in an oven maintained at 65°C (149°F) until the sample attains oven temperature. The bottle shall then be tightly capped and allowed to remain in the oven for 96 hours. At the end of that time the sample shall be removed from the oven and allowed to come to room temperature when the bottle will be opened and the soap inspected for rancid odor and separation.

4.3.5.4 Color. The Gardner Liquid Color Scale1/ consists of a set of 18 tubes of colored liquid. The test sample shall be put in a similar tube and its place on the scale determined visually.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Type I soap in one or five gallon quantiites as specified (see 6.2) shall be packaged in accordance with 5.1.1.1.1 or 5.1.1.1.2 as appropriate for the quantity.

5.1.1.1.1 One gallon quantity. One gallon of type I soap shall be furnished in an oblong, metal can with a bridge type handle conforming to PPP-C-96, type V, class 4.

5.1.1.1.2. Five gallon quantity. Five gallons of type I soap shall be furnished in a pail conforming to PPP-P-704, type I, class 3. The pail shall be closed in accordance with the appendix of the pail specification.

5.1.1.2 Type II. Type II soap of one class only in 35 or 120 pound quantities as specified (see 6.2) shall be packaged in accordance with 5.1.1.2.1 or 5.1.1.2.2 as appropriate for the quantity.

5.1.1.2.1 Thirty-five pound quantity. Thirty-five pounds of type II soap shall be furnished in a pail conforming to PPP-P-704, type II, class 3 or type III, class 3.

5.1.1.2.2 One-hundred and twenty pound quantity. One-hundred and twenty pounds of type II soap shall be furnished in a drum conforming to PPP-D-705 type III.

5.1.2 Level b.

5.1.2.1 Type I. Type I soap in one or five gallon quantities as specified (see 6.2) shall be packaged in accordance with 5.1.2.1.1 or 5.1.2.1.2 as appropriate for the quantity.

5.1.2.1.1 One gallon quantity. One gallon of type I soap shall be furnished in a can conforming to 5.1.1.1.1 or in a plastic bottle with a handle and closed to prevent leakage.

5.1.2.2 Type II. Type II soap of one class only in 35 or 120 pound quantities as specified (see 6.2) shall be packaged in accordance with 5.1.2.2.1 as appropriate for the quantity.

5.1.2.2.1 Thirty-five pound quantity. Thirty-five pounds to type II soap shall be furnished in a pail conforming to PPP-P-704, type II, class 1, 3, or 10 or type III, class 1, 3, or 5.

5.1.2.2.2 One-hundred and twenty pound quantity. One-hundred and twenty pounds of type II soap shall be furnished in a drum conforming to PPP-D-705, type III.

1/ Available from Gardner Labs Inc. 5521 Landy Lane, Bethesda, Maryland.

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5.1.3 Level C. The soap shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. The one-gallon metal cans shall be packed in accordance with level A packing requirements specified in the appendix of PPP-C-96. The five gallon quantity packaged as specified in 5.1.1.1.2 does not require additional packing.

5.2.1.2 Type II. The thirty-five pound pail and one-hundred and twenty pound packaged as specified in 5.1.2.2.1 and 5.1.2.2.2 do not require additional packing.

5.2.2 Level B.

5.2.2.1 Type I. Six (6) one-gallon containers shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be provided pads, partitions or dividers fabricated of fiberboard conforming to PPP-F-320, grade 200. The box shall be closed in accordance with Method II of the appendix of the box specification.

5.2.2.2 Type II. The thirty-five pound pail and the one-hundred and twenty pound drum packaged as specified in 5.1.2.2.1 and 5.1.2.2.2 do not require additional packing.

5.2.3 Level C. One type and class of soap in quantities as specified (see 6.2) shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with Uniform Freight or the National Motor Classification, as applicable.

6. NOTES

6.1 Intended use. Soaps of type I and type II are intended primarily for use on all kinds of floors, except on rubber flooring.

6.1.1 A solution of either 2 or 4 ounces of the liquid soap (type I) or 1 to 2 ounces of the paste soap (type II) per gallon of soft water should be satisfactory for cleaning operations. If the paste soap is not all dissolved, there will be waste and some undissolved soap may be left on the surface being cleaned. If the soap is all dissolved and the surface on which it is used is thoroughly rinsed with plain water, material conforming to type II should be satisfactory for cleaning operations. A ready-prepared liquid soap (type I) is time saving as it is in a condition to dilute and apply.

6.1.2 All ordinary soaps (liquid, or paste) when dissolved in hard water will deposit insoluble calcium and magnesium soaps. The removal of these insoluble soaps requires thorough rinsing or mopping with plain water.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of soap required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Quantities required (see 5.1.1.1, 5.1.1.2, 5.1.2.2 and 5.2.3).

6.3 Basis of purchase. Liquid soap (type I) should be purchased by volume. A gallon of soap contains 231 cubic inches at 15.5°C. (60°F.). Paste soap (type II) should be purchased by net weight.

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